

authentication. The trial account can be created for 30 days.

```
$ cf login -a api.ng.bluemix.net
```

Next, log on to the IBM container plugin so that the Docker images can be programmed and accessed in IBM Bluemix, as follows:

```
$ cf ic login
```

The name space on the IBM Bluemix panel is specified in the next instruction:

```
$ cf ic build -t mynamespace/bluechatter ./
```

In the next step, verify if the images have been created with the push to the IBM Bluemix container image registry. The images in IBM Bluemix can be programmed effectively once they are visible in the Bluemix panel.

```
$ cf ic images
```

The BlueChatter app is integrated with Redis; so there is a need to create a new Redis service, which can store all the chat communication in the storage space, as follows:

```
$ cf create-service rediscloud 10mb redis-chatter
```

By following all these instructions, the Docker image can be deployed on the Real Cloud IBM Bluemix. In addition to the classical deployment, applications can be scaled for multiple users and access rights. The number of users can be set in the instruction, which can send and receive the messages using Redis services.

```
$ cf scale MyApp -i 10
```

The customisation of the app with the new code can be done in IBM Bluemix. Using this approach, the new features in the application can be integrated and deployed using IBM DevOps services (JazzHub).

```
$ git clone https://github.com/CodenameBlueMix/bluechatter.git
$ cd bluechatter
$ cf push MyApp
$ cf app MyApp
```

To evaluate the features and performance of the app deployed to IBM Bluemix, it can be tested on various Web browsers.

Kubernetes and cluster programming

Kubernetes or K8s (<https://kubernetes.io/>) is a free and open source distribution that automates the scaling, deployment and

management of container based applications. It was initially developed by Google and subsequently donated to the Cloud Native Computing Foundation.

To work with Kubernetes and IBM Bluemix, there is need to work on Kubernetes CLI, which is available on <https://kubernetes.io/docs/tasks/tools/install-kubectl/>.

```
bx login -a https://api.ng.bluemix.net
bx plugin install container-service -r Bluemix
```

For the initialisation of the container service plugin, the following command is executed:

```
bx cs init
```

To create a new cluster, the free-tier cluster in IBM Bluemix can be selected. By using the following command, a cluster that has a single node with 4GB memory and two vCPUs will be created:

```
bx cs cluster-create --name my-cluster
```

The clusters can be listed using the following command:

```
bx cs clusters
```

To access the dashboard of Bluemix Kubernetes, the following command is executed on the local system:

```
kubectl proxy
```

The IBM Bluemix Container Service with Kubernetes can be viewed on <http://127.0.0.1:8001/ui> using a Web browser.

Scope for R&D

There is huge scope for research activities and projects using Docker container technology. As NoSQL databases are now widely used for Big Data based applications, the integration of a NoSQL database, including MongoDB, Cassandra, CouchDB, Neo4J and many others, can be programmed and attached with the containers, and their performance can be evaluated. The distribution, access and sharing of database services with security on a real, hybrid cross-platform cloud can be programmed, along with the evaluation of assorted parameters including performance, privacy, security, integrity, cost factor, complexity and many others. 

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